

**LAPORAN PRAKTIKUM**  
**WORKSHOP SISTEM INFORMASI WEB SERVER**  
**(TIF330805)**  
**SEMESTER III**



**Acara 14**  
**Security, Exception, Logging, PRG, dan Integrasi CRUD Lengkap**

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**POLITEKNIK NEGERI JEMBER**  
**TAHUN 2026**

```

class MahasiswaRepository extends BaseRepository
{
    // JOIN ke prodi supaya daftar menampilkan nama prodi
    private const SELECT = "SELECT m.id, m.nim, m.nama, m.email,
m.prodi_id, p.nama AS prodi, m.angkatan, m.status
                        FROM mahasiswa m
                        JOIN prodi p ON p.id = m.prodi_id";

    /** @return Mahasiswa[] */
    public function all(): array
    {
        $stmt = $this->pdo->prepare(self::SELECT . " ORDER BY
m.nim");
        $stmt->execute();

        return array_map([Mahasiswa::class, 'fromRow'], $stmt-
>fetchAll());
    }
}

```

1. MahasiswaRepository menggunakan prepared statement

```

// ----- Validasi (private) -----
private function normalize(array $in): array
{
    return [
        'id'      => (int) ($in['id'] ?? 0),
        'nim'     => trim((string) ($in['nim'] ?? '')),
        'nama'    => trim((string) ($in['nama'] ?? '')),
        'email'   => trim((string) ($in['email'] ?? '')),
        'prodi_id' => (int) ($in['prodi_id'] ?? 0),
        'angkatan' => (int) ($in['angkatan'] ?? date('Y')),
        'status'  => (string) ($in['status'] ?? 'aktif'),
    ];
}

// Isi entity lewat setter; setter yang gagal validasi
dikumpulkan sebagai error
// (validasi NIM, nama, email, prodi, angkatan, status)
private function fill(Mahasiswa $mhs, array $input): array
{
    $errors = [];

    foreach ($input as $field => $value) {
        $setter = 'set' . str_replace('_', '',
ucwords($field, '_')); // prodi_id => setProdiId
        try {

```

```

        $mhs->$setter($value);
    } catch (InvalidArgumentException $e) {
        $errors[] = $e->getMessage();
    }
}

```

2. Service dengan validasi dan Business Logic.

```

catch (PDOException $e) {
    // Dua request bersamaan dengan NIM sama: ditolak
    oleh UNIQUE di database
    if ($this->isDuplicateKey($e)) {
        return $this->fail($input, [], 'NIM sudah
    terdaftar.');
```

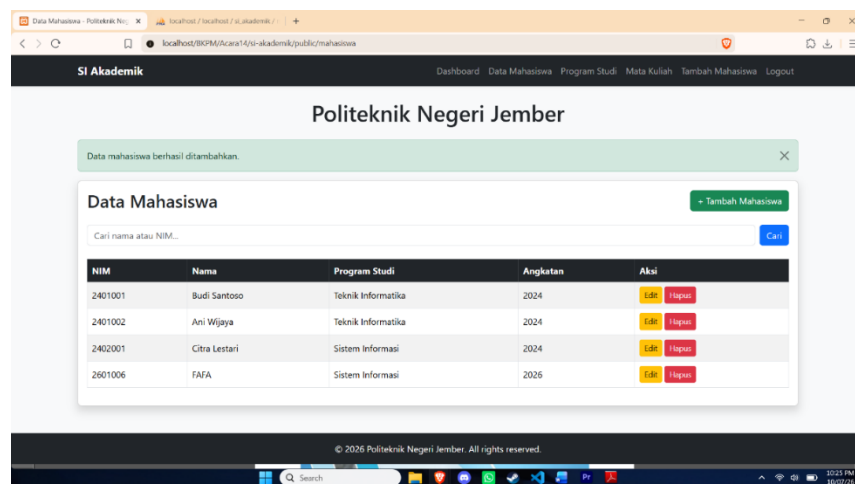
3. Exception Handling.

```

app.log U X
Acara14 > si-akademik > storage > logs > app.log
1 [2026-10-07 15:30:42] ERROR: Gagal menambah mahasiswa |
2 PDOException: SQLSTATE[42S02]: Base table or view not found: 1146
3 Table 'si_akademik.mahasiswa_x' doesn't exist (MahasiswaRepository.php:59)
4 | {"nim":"2401006"}
5

```

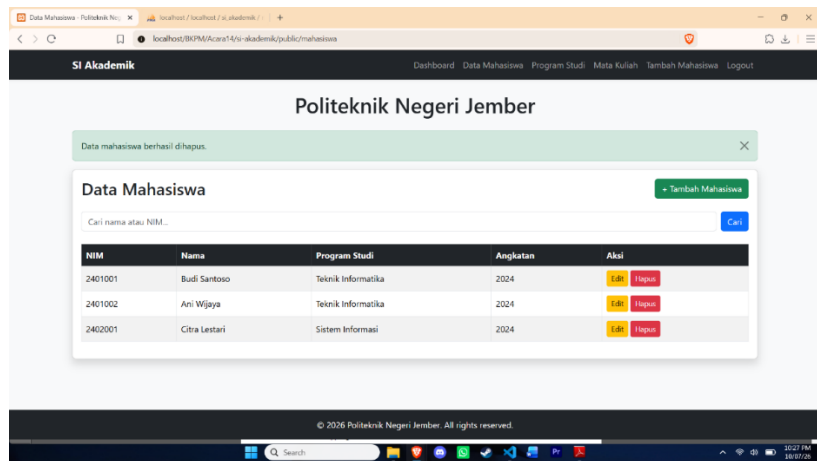
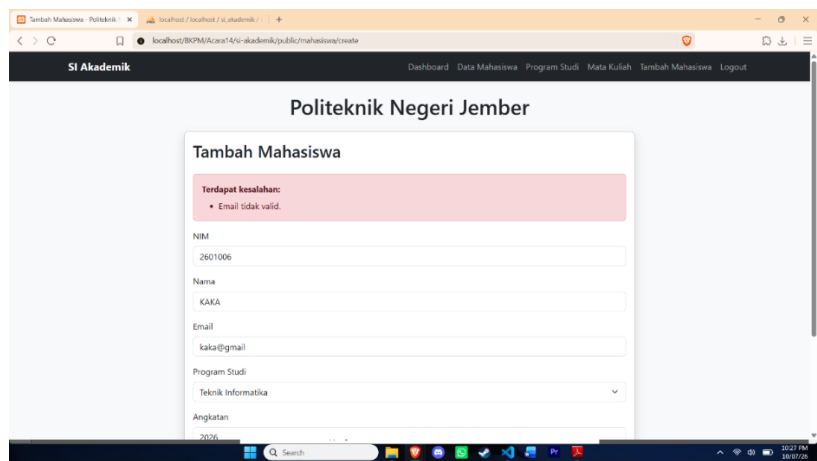
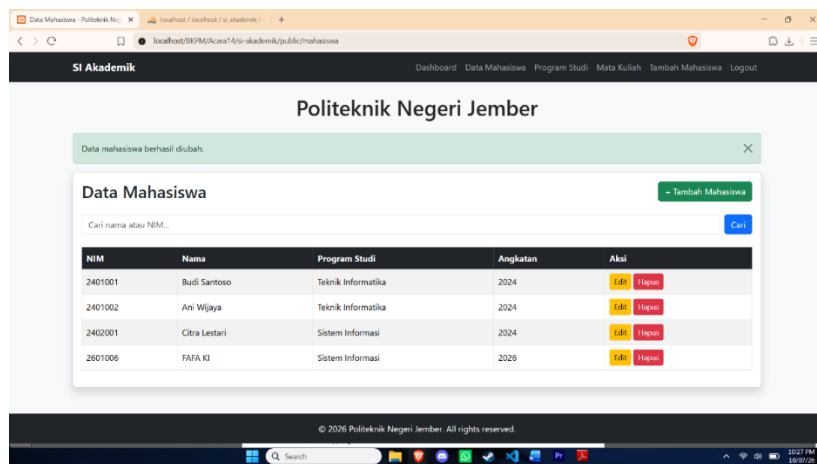
4. File app.log yang sudah dicoba menggunakan metode simulasi error database

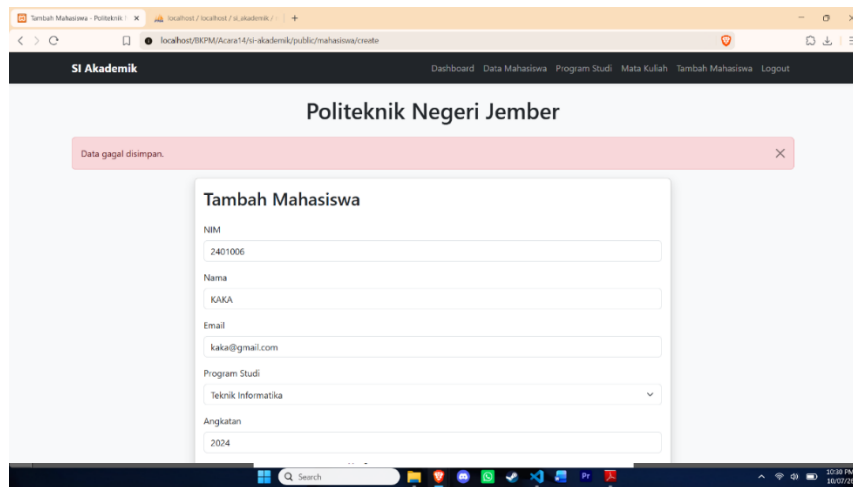


The screenshot displays a web application for Politeknik Negeri Jember. At the top, there's a navigation bar with links like 'Dashboard', 'Data Mahasiswa', 'Program Studi', 'Mata Kuliah', 'Tambah Mahasiswa', and 'Logout'. Below this, a green notification banner states 'Data mahasiswa berhasil ditambahkan.' The main content area is titled 'Data Mahasiswa' and features a search bar with the placeholder 'Cari nama atau NIM...'. Below the search bar is a table with the following data:

| NIM     | Nama          | Program Studi      | Angkatan | Aksi                                       |
|---------|---------------|--------------------|----------|--------------------------------------------|
| 2401001 | Budi Santoso  | Teknik Informatika | 2024     | <a href="#">Edit</a> <a href="#">Hapus</a> |
| 2401002 | Ani Wijaya    | Teknik Informatika | 2024     | <a href="#">Edit</a> <a href="#">Hapus</a> |
| 2402001 | Citra Lestari | Sistem Informasi   | 2024     | <a href="#">Edit</a> <a href="#">Hapus</a> |
| 2601006 | FAFA          | Sistem Informasi   | 2026     | <a href="#">Edit</a> <a href="#">Hapus</a> |

At the bottom of the page, there is a footer that reads '© 2026 Politeknik Negeri Jember. All rights reserved.'





## 5. Flash Message